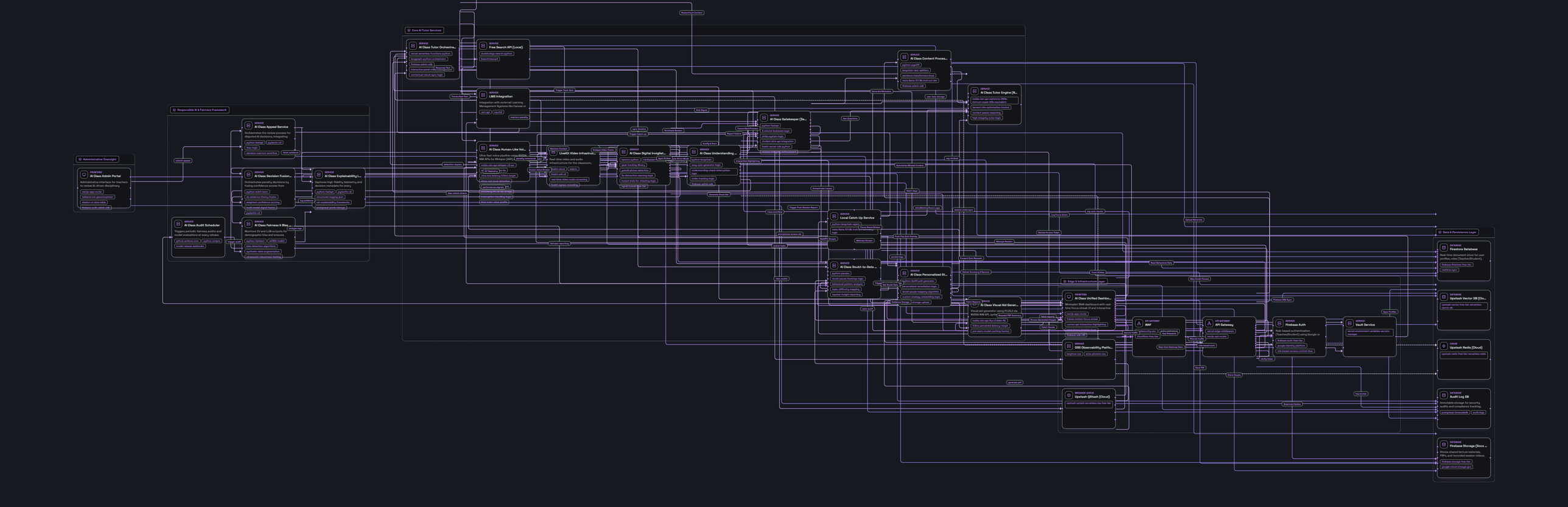

Administrative Oversight
FRONTEND
AI Class Admin Portal
Administrative interface for teachers to review AI-driven disciplinary
nextjs-app-router
tailwind-css-glassmorphism
shadcn-ui-data-table
firebase-auth-admin-sdk
Responsible AI & Fairness Framework
SERVICE
AI Class Appeal Service
Orchestrates the review process for disputed AI decisions, integrating
python-fastapi
pydantic-v2
rbac-logic
decision-overturn-workflow
SERVICE
AI Class Decision Fusion...
Orchestrates penalty decisions by fusing confidence scores from
python-scikit-learn
ds-evidence-theory-fusion
weighted-confidence-scoring
multi-modal-signal-fusion
pydantic-v2
SERVICE
AI Class Explainability L...
Captures high-fidelity telemetry and decision metadata for every
python-fastapi
pydantic-v2
structured-logging-json
xai-explainability-frameworks
postgresql-jsonb-storage
SERVICE
AI Class Audit Scheduler
Triggers periodic fairness audits and model evaluations at every release.
github-actions-cron
python-scripts
model-release-webhooks
SERVICE
AI Class Fairness & Bias ...
Monitors CV and LLM outputs for demographic bias and ensures
python-fairlearn
aif360-toolkit
bias-detection-algorithms
synthetic-data-augmentation
adversarial-robustness-testing
submit-appeal
fetch-evidence
log-evidence
trigger-audit
analyze-logs
Core AI Tutor Services
SERVICE
AI Class Tutor Orchestra...
vercel-serverless-functions-python
langgraph-python-orchestrator
firebase-admin-sdk
interactive-panel-state-management
contextual-visual-sync-logic
Response Text
SERVICE
Free Search API (Local)
duckduckgo-search-python
beautifulsoup4
SERVICE
LMS Integration
Integration with external Learning Management Systems like Canvas or
rest-api
oauth2
overturn-penalty
Transcribed Text
SERVICE
AI Class Human-Like Voi...
Ultra-fast voice pipeline using NVIDIA NIM APIs for Whisper (ASR) a
nvidia-nim-api-whisper-v3-asr
silero-vad-local-detection
streaming-llm-to-tts-bridge
interruption-handling-logic
kind-tutor-voice-profile
ultra-low-latency-0.5ms-target
detection-signals
penalty-commands
CV Telemetry
performance-signals
Human Voice Stream
bias-check-stream
SERVICE
LiveKit Video Infrastruc...
Real-time video and audio infrastructure for the classroom,
livekit-cloud
webrtc
livekit-sdk-v2
real-time-video-audio-streaming
livekit-egress-recording
Decision Context
SERVICE
AI Class Digital Invigilat...
opencv-python
gaze-tracking-library
yolov8-phone-detection
3s-distraction-warning-logic
instant-kick-for-cheating-logic
ngrok-tunnel-free-tier
Analyze Video Tracks
Sync Strikes
SERVICE
AI Class Understanding ...
python-langchain
easy-quiz-generator-logic
understanding-check-interruption-logic
strike-tracking-logic
firebase-admin-sdk
Config & Start
Interactive Highlighting
SERVICE
AI Class Gatekeeper (Se...
python-fastapi
3-minute-lockdown-logic
strike-system-logic
student-kick-api-integration
livekit-server-sdk-python
session-integrity-enforcement
Reasoning & Content
Trigger Topic Quiz
Kick Signal
sync timeline
Trigger Catch-up
Terminate Session
Transcribe & Process
Report Failure
Gen Questions
Generate Metadata
user data storage
Generate Visual Aid
Orchestrate Lesson
Trigger Post-Session Report
trace workflow
microphone-stream-stt
queue tasks
fairness-reporting
Summarize Missed Content
rag retrieval
fetch-keys
Join/Waiting Room Logic
session state sync
SERVICE
AI Class Content Proces...
python-pypdf2
langchain-text-splitters
sentence-transformers-local
meta-llama-3.1-8b-instruct-nim
firebase-admin-sdk
SERVICE
AI Class Tutor Engine (N...
nvidia-nim-api-nemotron-340b-instruct-super-49b-equivalent
tensorrt-llm-optimization-hosted
context-aware-reasoning
high-integrity-tutor-logic
SERVICE
Local Catch-Up Service
python-langchain-agent
meta-llama-3.1-8b-instruct-summary-logic
Focus Score Stream
WebRTC Stream
Webcam Stream
Push Pop Quiz Overlay
persist-logs
Forward Quiz Request
Deliver Summary & Resume
Log Focus Score
log-quiz-results
Revoke Access Token
Manage Session
Upload Materials
SERVICE
AI Class Doubt-to-Data ...
python-pandas
doubt-pause-heatmap-logic
behavioral-pattern-analysis
topic-difficulty-mapping
teacher-insight-reporting
view-audits
Get Doubt Data
Evidence Storage
save-audit
SERVICE
AI Class Personalized St...
python-fpdf2-pdf-generator
personalized-remediation-logic
doubt-pause-mapping-algorithm
custom-analogy-embedding-logic
SERVICE
AI Class Visual Aid Gener...
Visual aid generator using FLUX.2 via NVIDIA NIM API, optim
nvidia-nim-api-flux-2-klein-4b
0.5ms-perceived-latency-target
pre-warm-model-caching-hosted
Fetch Reports
Request PDF Summary
fetch reports
Stream Generated Images
Fetch Visuals
generate pdf
Read Behavioral Data
Track Strikes
Map Doubt Pauses
Edge & Infrastructure Layer
FRONTEND
AI Class Unified Dashbo...
Minimalist B&W dashboard with real-time focus streak UI and interactive
nextjs-app-router
framer-motion-focus-streak
canvas-api-interactive-highlighting
react-konva-drawing-layer
firebase-sdk-v10
SERVICE
OSS Observability Platfo...
langfuse-oss
arize-phoenix-oss
MESSAGE-QUEUE
Upstash QStash (Cloud)
upstash-qstash-serverless-mq-free-tier
API GATEWAY
WAF
nginx-waf-local
cloudflare-free-tier
secure-traffic
App Requests
Real-time Heatmap Data
API GATEWAY
API Gateway
vercel-edge-middleware
nextjs-api-routes
filtered-traffic
role-based auth
Firebase SDK Sync
SERVICE
Firebase Auth
Role-based authentication (Teacher/Student) using Google or
firebase-auth-free-tier
google-identity-platform
role-based-access-control-rbac
verify-token
Save PDF
Store Visuals
log-access
Download Guides
SERVICE
Vault Service
vercel-environment-variables-secrets-manager
Sync Profiles
Data & Persistence Layer
DATABASE
Firestore Database
Real-time document store for user profiles, roles (Teacher/Student),
firebase-firestore-free-tier
realtime-sync
DATABASE
Upstash Vector DB (Clo...
upstash-vector-free-tier-serverless-vector-db
CACHE
Upstash Redis (Cloud)
upstash-redis-free-tier-serverless-redis
DATABASE
Audit Log DB
Immutable storage for security audits and compliance tracking.
postgresql-timescaledb
audit-logs
DATABASE
Firebase Storage (Docs ...
Stores shared lecture materials, PDFs, and recorded session videos.
firebase-storage-free-tier
google-cloud-storage-gcs